

ADARSH KUMAR

AI/ML Engineer

+91 9801742363 | adarshkumarsbrhs@gmail.com | LinkedIn | Github

PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience in **Generative AI, NLP, RAG, LLMs**, and **Multi-Agent Systems** through a Data Science Internship at Gatim AI Tech Innovation Pvt Ltd and independent AI projects. Built healthcare chatbots, conversational AI platforms, and AI research copilots using **Python, FastAPI, LangChain, MongoDB, Redis**, and **Groq**. Skilled in developing end-to-end AI applications spanning retrieval pipelines, agentic workflows, multimodal interactions, and scalable backend systems.

TECHNICAL SKILLS

- **Backend Languages:** Python, Java, TypeScript, SQL
- **Frameworks & Libraries:** FastAPI, Hugging Face, PyTorch, Scikit-learn, Pandas, NumPy, Seaborn, Matplotlib, LangChain, LangGraph, MCP, A2A, ChromaDB, CrewAI, Phidata, Google adk
- **AI/ML & GenAI:** Machine Learning, Deep Learning, Generative AI, Agentic AI, NLP, RAG, LLMs, Prompt Engineering
- **Data Science:** EDA, Feature Engineering, Data Cleaning, Model Evaluation, Descriptive Statistics, Visualization
- **Databases:** MongoDB, Redis, Supabase
- **DevOps & Tools:** Git/GitHub, Docker, CI/CD
- **Additional:** RESTful APIs, JWT Authentication, OAuth 2.0, WebSockets, Socket.io, Kafka, RabbitMQ, Microservices Architecture, OpenAPI, GroqAPI

PROFESSIONAL EXPERIENCE

Data Science Intern, Gatim AI Tech Innovation Pvt Ltd

Jul 2024 - Dec 2024 | Remote

- **Applied Generative AI** and **NLP** techniques to real-world healthcare datasets, performing data preparation and model optimization that strengthened multilingual information retrieval and overall chatbot response quality.
- **Developed** a **RAG-based** healthcare chatbot with multilingual support, voice-to-voice interaction, and image-based question answering, enabling users to access healthcare information through multiple input and output formats.
- **Delivered** real-time question-answering and translation features using **LangChain, Meta LLaMA, Groq API**, and **Hugging Face**, enhancing chatbot functionality and supporting multilingual healthcare communication.
- **Optimized** text preprocessing, embedding generation, and **vectorization** techniques to improve information retrieval performance, increasing answer relevance by 30% during project evaluation and testing.

PROJECTS

BrainWeave ARC | Python, FastAPI, MongoDB, Redis, FAISS, Docker

Dec 2025 – Apr 2026

- **Engineered** BrainWeave ARC, a multi-agent AI research copilot using **Python, FastAPI, TypeScript, MongoDB, Redis**, and **Docker**, featuring seven specialized agents for planning, research, coding, validation, repair, and report generation.
- **Designed** an end-to-end **RAG pipeline** with PDF/DOCX ingestion, semantic chunking, **Sentence Transformers** embeddings, and **FAISS** vector indexing, enabling top-K retrieval across 100+ document chunks for context-aware, citation-backed responses.
- **Integrated Groq LLaMA-3.3-70B** to power multi-agent reasoning, automated research synthesis, code generation, and fact validation, while implementing **Redis** caching that cut repeated LLM requests and reduced inference cost by 25%.
- **Implemented** real-time communication via **WebSockets, JWT authentication**, and **RBAC**, delivering sub-200ms progress updates for secure multi-user access with one-click export to professionally formatted PDF and DOCX reports.

Neuro | React.js, Python, FastAPI, MongoDB, LangChain, Groq, Hugging Face

May 2025 – Oct 2025

- **Built** Neuro Chat AI, a multilingual conversational platform using **React.js, Express.js, FastAPI**, and **MongoDB**, featuring 9 specialized AI chatbots supporting real-time text and voice interaction for diverse user needs.
- **Implemented Speech-to-Text (STT)** and **Text-to-Speech (TTS)** pipelines enabling real-time voice conversations, improving platform accessibility and reducing language-barrier friction by an estimated 30% across multilingual user sessions.
- **Developed** secure dual-mode authentication using facial recognition and email-based login, encrypting biometric embeddings and user credentials to prevent unauthorized access across 100% of active user sessions.
- **Integrated LangChain, Groq**, and **Hugging Face** models to enable context-aware conversations with persistent memory and sub-second AI inference, storing full chat history in **MongoDB** via a scalable REST API architecture.

EDUCATION

Bachelor of Technology in Computer Engineering

Aug 2022 – May 2026

RK University, Rajkot, Gujarat

LEADERSHIP & COMMUNITY ENGAGEMENT

Treasurer – NEURON (AI & ML Club)

Mar 2024 – May 2026

RK University

Rajkot, Gujarat

- Conducted technical workshops on building production-ready backend systems, API development, and database architecture for 100+ students.
- Led hands-on sessions covering REST API design, authentication strategies, microservices patterns, query optimization, and deployment best practices.